PHYDSC401T — Mathematical Physics-III

Unit 5: Group Theory and Numerical Methods

7th Semester, Assam University (FYUG) — Haflong Government College

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Notes 7 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

1. Abstract groups: subgroups, classes, cosets, factor groups, normal subgroups, direct product of groups; Homomorphism and Isomorphism.
2. Solution of algebraic and transcendental equations by Bisection and Newton Raphson method.
3. Solution of 1st order differential equation using Euler and 2nd order Runge-Kutta method.

Top 2 high-mark topics in this unit

1. Newton-Raphson and Bisection methods for algebraic and transcendental equations — 2 of the 3 AU questions in this unit are 4-mark Sec B questions on bisection and Newton-Raphson root finding.
2. Abstract groups: subgroups, cosets, normal subgroups, factor groups, homomorphism and isomorphism — half the unit by syllabus, with zero AU questions on group theory, so it is the biggest untested gap and a definition plus proof topic that can carry long answers.

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1. Groups: Definition and Examples

Definition of a Group

A group is an algebraic structure consisting of a non-empty set G together with a binary operation $\cdot$ (often referred to as multiplication or group addition) that maps any ordered pair $(a, b) \in G \times G$ to an element $a \cdot b \in G$. For the pair $(G, \cdot)$ to form a group, the operation must satisfy four fundamental axioms:

1. **Closure**: For all elements $a, b \in G$, the product $a \cdot b$ belongs to G.

$$a \cdot b \in G, \quad \forall a, b \in G$$

1. **Associativity**: For all elements $a, b, c \in G$, the order of operation evaluation does not affect the result.

$$(a \cdot b) \cdot c = a \cdot (b \cdot c), \quad \forall a, b, c \in G$$

1. **Identity Element**: There exists an element $e \in G$, called the identity element, such that operating with any element $a \in G$ leaves the element unchanged.

$$a \cdot e = e \cdot a = a, \quad \forall a \in G$$

1. **Inverse Element**: For every element $a \in G$, there exists an element $a^{-1} \in G$, called the inverse of a, such that operating a with a^{-1} yields the identity element.

$$a \cdot a^{-1} = a^{-1} \cdot a = e$$

If a group additionally satisfies the commutativity condition $a \cdot b = b \cdot a$ for all $a, b \in G$, it is called an **Abelian group** (or commutative group). If commutativity does not hold for all elements, the group is called **non-Abelian**. The number of elements in a group G is called the **order** of the group, denoted by $|G|$ or $o(G)$. If the number of elements is finite, G is a finite group; otherwise, it is an infinite group.

Examples of Groups

1. Integers under Addition

Consider the set of integers $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ with the binary operation of standard addition $+$.

- **Closure**: For any two integers $a, b \in \mathbb{Z}$, their sum $a + b$ is always an integer.

$$a + b \in \mathbb{Z}$$

- **Associativity**: Addition of integers is associative for all $a, b, c \in \mathbb{Z}$.

$$(a + b) + c = a + (b + c)$$

- **Identity Element**: The integer $0 \in \mathbb{Z}$ acts as the additive identity because for any $a \in \mathbb{Z}$:

$$a + 0 = 0 + a = a$$

- **Inverse Element**: For every integer $a \in \mathbb{Z}$, its additive inverse is $-a \in \mathbb{Z}$, satisfying:

$$a + (-a) = (-a) + a = 0$$

Furthermore, since $a + b = b + a$ for all $a, b \in \mathbb{Z}$, the structure $(\mathbb{Z}, +)$ forms an infinite Abelian group.

2. Non-zero Real Numbers under Multiplication

Consider the set of non-zero real numbers $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ under standard multiplication $\cdot$.

- **Closure**: The product of any two non-zero real numbers $a, b \in \mathbb{R}^*$ is non-zero, so $a \cdot b \in \mathbb{R}^*$.

$$a \cdot b \neq 0$$

- **Associativity**: Multiplication of real numbers is associative for all $a, b, c \in \mathbb{R}^*$.

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

- **Identity Element**: The real number $1 \in \mathbb{R}^*$ is the multiplicative identity.

$$a \cdot 1 = 1 \cdot a = a$$

- **Inverse Element**: For every non-zero real number $a \in \mathbb{R}^*$, the inverse element is its reciprocal $a^{-1} = \frac{1}{a} \in \mathbb{R}^*$, giving:

$$a \cdot \frac{1}{a} = \frac{1}{a} \cdot a = 1$$

The set $(\mathbb{R}^*, \cdot)$ is an infinite Abelian group. Notice that the entire set of real numbers $\mathbb{R}$ does not form a group under multiplication because 0 has no multiplicative inverse.

3. Symmetry Group of an Equilateral Triangle

The symmetry group of an equilateral triangle, denoted D_3 (or C_{3v} in physical notation), consists of all spatial transformations that map the triangle onto itself. These transformations include three rotations about its center and three reflections across its axes of symmetry.

Let the three vertices of the triangle be labeled $1, 2, 3$. The set of transformations comprises six operations: - e: Identity transformation (rotation by 0°). - r: Counter-clockwise rotation by 120° ($2\pi/3$). - r^2: Counter-clockwise rotation by 240° ($4\pi/3$). - s_1, s_2, s_3: Reflections across the lines passing through vertices $1, 2, 3$ and the midpoints of their opposite sides, respectively.

Alternatively, expressing all reflections in terms of a single fundamental reflection $s = s_1$, the elements can be written as:

$$D_3 = \{e, r, r^2, s, sr, sr^2\}$$

The group operation is the sequential composition of transformations. The operational relations defining this group are:

$$r^3 = e$$

$$s^2 = e$$

$$sr = r^{-1}s = r^2s$$

The composition of transformations is associative, the identity transformation e leaves all vertices invariant, and every transformation has a unique reverse transformation. Because $sr = r^2s \neq rs$, D_3 is a non-Abelian group of order $|D_3| = 6$.

4. Permutation Group S_3

The symmetric group S_3 is the set of all bijective functions (permutations) from the set $\{1, 2, 3\}$ to itself, under the binary operation of function composition.

A permutation σ mapping $1 \mapsto \sigma(1), 2 \mapsto \sigma(2), 3 \mapsto \sigma(3)$ can be written in two-line notation or cycle notation. The $3! = 6$ distinct elements of S_3 in cycle notation are: - Identity permutation: (1) - Transpositions (2-cycles): $(12), (23), (13)$ - 3-cycles: $(123), (132)$

Thus, the set is:

$$S_3 = \{(1), (12), (23), (13), (123), (132)\}$$

Composition of permutations proceeds from right to left. For example, evaluating $(12)(123)$: The element 1 under (123) goes to 2, and 2 under (12) goes to 1.

$$(12)(123)(1) = 1$$

The element 2 under (123) goes to 3, and 3 under (12) goes to 3.

$$(12)(123)(2) = 3$$

The element 3 under (123) goes to 1, and 1 under (12) goes to 2.

$$(12)(123)(3) = 2$$

Combining these mappings yields the cycle (23):

$$(12)(123) = (23)$$

Function composition is associative, the identity permutation leaves all elements fixed, and every bijection has an inverse bijection. The group S_3 is non-Abelian (for instance, $(12)(23) = (123)$ whereas $(23)(12) = (132)$) and has order $|S_3| = 6$. It is isomorphic to the symmetry group of an equilateral triangle D_3.

2. Subgroups

Definition of a Subgroup

Let $(G, \cdot)$ be a group. A non-empty subset $H \subseteq G$ is called a **subgroup** of G if H itself forms a group under the restricted operation $\cdot$ of G. We denote H being a subgroup of G by:

$$H \leq G$$

If $H \leq G$ and $H \neq G$, then H is a proper subgroup, written as $H < G$. Every group G containing at least two elements has at least two trivial subgroups: the identity subgroup $\{e\}$ and G itself.

Subgroup Tests

To verify whether a non-empty subset $H \subseteq G$ is a subgroup, it is not necessary to re-verify all four group axioms individually. We can use simplified subgroup criteria.

Two-Step Subgroup Test

A non-empty subset H of a group G is a subgroup of G if and only if:

1. H is closed under the group operation:

$$a, b \in H \implies a \cdot b \in H$$

1. H is closed under taking inverses:

$$a \in H \implies a^{-1} \in H$$

One-Step Subgroup Test

A non-empty subset H of a group G is a subgroup of G if and only if for all $a, b \in H$, the element $a \cdot b^{-1}$ belongs to H.

$$a, b \in H \implies a \cdot b^{-1} \in H$$

Proof of the One-Step Subgroup Test Assume $H \subseteq G$ is non-empty and satisfies $a \cdot b^{-1} \in H$ for all $a, b \in H$. We verify the group requirements for H:

Since H is non-empty, choose an arbitrary element $x \in H$. Setting $a = x$ and $b = x$ in the condition gives:

$$x \cdot x^{-1} \in H$$

Because $x \cdot x^{-1} = e$ in G, we find:

$$e \in H$$

This shows that H contains the identity element.

Next, for any $x \in H$, using $a = e \in H$ and $b = x \in H$ in the condition gives:

$$e \cdot x^{-1} \in H$$

Since $e \cdot x^{-1} = x^{-1}$, we obtain:

$$x^{-1} \in H$$

Thus, H is closed under inverses.

Finally, let $x, y \in H$. Since $y \in H$, its inverse $y^{-1} \in H$. Setting $a = x$ and $b = y^{-1}$ in the condition gives:

$$x \cdot (y^{-1})^{-1} \in H$$

Since $(y^{-1})^{-1} = y$, this reduces to:

$$x \cdot y \in H$$

Thus, H is closed under the operation. Associativity holds automatically in H because it holds for all elements in G. Therefore, $H \leq G$.

Examples of Subgroups

1. **Even Integers in $(\mathbb{Z}, +)$**:

Consider the subset of even integers $2\mathbb{Z} = \{\dots, -4, -2, 0, 2, 4, \dots\} \subset \mathbb{Z}$. Let $a, b \in 2\mathbb{Z}$. Then $a = 2k$ and $b = 2m$ for some $k, m \in \mathbb{Z}$. Applying the additive one-step test:

$$a - b = 2k - 2m = 2(k - m)$$

Since $k - m \in \mathbb{Z}$, $a - b \in 2\mathbb{Z}$. Thus, $2\mathbb{Z} \leq \mathbb{Z}$.

1. **Rotations in S_3**:

Consider the subset of rotational symmetries $A_3 = \{(1), (123), (132)\} \subset S_3$. The product of any two 3-cycles or identity yields an element in A_3, and the inverse of (123) is $(132) \in A_3$. Thus, $A_3 \leq S_3$.

Cyclic Subgroups

Let a be an element of a group G. The set of all integer powers of a, defined as:

$$\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$$

forms a subgroup of G. This subgroup $\langle a \rangle$ is called the **cyclic subgroup generated by** a.

Proof that $\langle a \rangle$ is a Subgroup

The set $\langle a \rangle$ is non-empty because $a^1 = a \in \langle a \rangle$. Let $x, y \in \langle a \rangle$. Then $x = a^r$ and $y = a^s$ for some integers $r, s \in \mathbb{Z}$. The inverse of y is given by:

$$y^{-1} = (a^s)^{-1} = a^{-s}$$

Now form the product $x \cdot y^{-1}$:

$$x \cdot y^{-1} = a^r \cdot a^{-s} = a^{r-s}$$

Since $r - s \in \mathbb{Z}$, it follows that $x \cdot y^{-1} \in \langle a \rangle$. By the one-step subgroup test, $\langle a \rangle \leq G$.

Order of an Element

The **order of an element** $a \in G$, denoted $o(a)$ or $|a|$, is the smallest positive integer n such that:

$$a^n = e$$

If no such positive integer exists, a is said to have infinite order. For any element $a \in G$, the order of the element equals the order of the cyclic subgroup generated by it:

$$o(a) = |\langle a \rangle|$$

3. Cosets and Lagrange's Theorem

Left and Right Cosets

Let H be a subgroup of a group G, and let a be an element of G.

- The **left coset** of H in G determined by a is the set:

$$aH = \{ah \mid h \in H\}$$

- The **right coset** of H in G determined by a is the set:

$$Ha = \{ha \mid h \in H\}$$

The element a is called the **coset representative**. If G is an Abelian group under multiplication, $aH = Ha$ for all $a \in G$. For a group under addition $+$, the left coset is written as $a + H = \{a + h \mid h \in H\}$.

Example of Left and Right Cosets in a Non-Abelian Group

Consider the symmetric group $S_3 = \{(1), (12), (23), (13), (123), (132)\}$ and its subgroup $H = \{(1), (12)\}$.

Let us calculate the left cosets of H in S_3:

$$(1)H = \{(1)(1), (1)(12)\} = \{(1), (12)\}$$

$$(13)H = \{(13)(1), (13)(12)\} = \{(13), (123)\}$$

$$(23)H = \{(23)(1), (23)(12)\} = \{(23), (132)\}$$

Now let us calculate the right cosets of H in S_3:

$$H(1) = \{(1)(1), (12)(1)\} = \{(1), (12)\}$$

$$H(13) = \{(1)(13), (12)(13)\} = \{(13), (132)\}$$

$$H(23) = \{(1)(23), (12)(23)\} = \{(23), (123)\}$$

Notice that $(13)H = \{(13), (123)\}$ while $H(13) = \{(13), (132)\}$. Thus, $(13)H \neq H(13)$, showing that left and right cosets of a subgroup need not be identical.

Properties of Cosets

Let H be a subgroup of G, and let $a, b \in G$. The left cosets of H satisfy the following foundational properties:

1. **Element Inclusion**: Every element $a \in G$ belongs to its own left coset aH.

$$a \in aH$$

Proof: Since $H \leq G$, the identity $e \in H$. Therefore, $a = ae \in aH$.

1. **Equivalence and Equality Criterion**: Two left cosets aH and bH are equal if and only if $a^{-1}b \in H$.

$$aH = bH \iff a^{-1}b \in H$$

1. **Partitioning Property**: Two left cosets aH and bH are either identical or completely disjoint. That is, if $aH \cap bH \neq \emptyset$, then $aH = bH$.

Proof: Suppose $c \in aH \cap bH$. Then there exist $h_1, h_2 \in H$ such that:

$$c = ah_1$$

$$c = bh_2$$

Equating the two expressions for c:

$$ah_1 = bh_2$$

Multiplying both sides on the right by h_1^{-1}:

$$ah_1h_1^{-1} = bh_2h_1^{-1}$$

$$a = b(h_2h_1^{-1})$$

Since $h_1, h_2 \in H$, the element $h_3 = h_2h_1^{-1} \in H$. Thus:

$$a = bh_3$$

Now for any arbitrary element $x \in aH$, there exists $h \in H$ such that $x = ah$. Substituting $a = bh_3$:

$$x = (bh_3)h = b(h_3h)$$

Since $h_3h \in H$, we have $x \in bH$, proving $aH \subseteq bH$. By symmetrical argument, $bH \subseteq aH$. Hence:

$$aH = bH$$

Because every $a \in G$ belongs to some left coset (specifically $a \in aH$), and any two distinct left cosets are disjoint, the collection of distinct left cosets forms a **partition** of the group G:

$$G = \bigcup_{a \in G} aH$$

1. **Equal Cardinality**: Every left coset aH has precisely the same number of elements as the subgroup H.

$$|aH| = |H|$$

Proof: Define a mapping $\phi : H \to aH$ by:

$$\phi(h) = ah, \quad \forall h \in H$$

To prove ϕ is injective (one-to-one), suppose $\phi(h_1) = \phi(h_2)$ for $h_1, h_2 \in H$:

$$ah_1 = ah_2$$

Multiplying on the left by a^{-1}:

$$a^{-1}ah_1 = a^{-1}ah_2$$

$$eh_1 = eh_2$$

$$h_1 = h_2$$

To prove ϕ is surjective (onto), take any element $y \in aH$. By definition of aH, $y = ah$ for some $h \in H$. Then:

$$\phi(h) = ah = y$$

Thus ϕ is a bijection, which proves $|aH| = |H|$.

Lagrange's Theorem

Statement

If G is a finite group and H is a subgroup of G, then the order of H divides the order of G.

$$|H| \text{ divides } |G|$$

Proof

Let G be a finite group of order n, so $|G| = n$. Let H be a subgroup of G of order m, so $|H| = m$.

Because G is finite, the number of distinct left cosets of H in G must be finite. Let k be the number of distinct left cosets of H in G. Denote these distinct left cosets by:

$$a_1H, a_2H, \ldots, a_kH$$

Since the distinct left cosets of H form a partition of G, G is the disjoint union of these k cosets:

$$G = a_1H \cup a_2H \cup \cdots \cup a_kH$$

Because the cosets are mutually disjoint, the number of elements in G is the sum of the number of elements in each distinct coset:

$$|G| = |a_1H| + |a_2H| + \cdots + |a_kH|$$

By the cardinality property of cosets, every left coset contains exactly $|H| = m$ elements:

$$|a_iH| = |H| = m, \quad \forall i = 1, 2, \ldots, k$$

Substituting these sizes into the sum equation:

$$|G| = \underbrace{m + m + \cdots + m}_{k \text{ terms}}$$

$$|G| = k \cdot m$$

Substituting $|G| = n$ and $|H| = m$:

$$n = k \cdot |H|$$

Rearranging for k:

$$k = \frac{|G|}{|H|}$$

Since k (the number of cosets) is an integer, $|H|$ divides $|G|$ exactly. This completes the proof.

Index of a Subgroup

The **index** of a subgroup H in a group G, denoted by $[G : H]$, is defined as the number of distinct left (or right) cosets of H in G.

For a finite group G, Lagrange's theorem provides a direct algebraic formula for the index:

$$[G : H] = \frac{|G|}{|H|}$$

Example

In the symmetric group S_3 of order $|S_3| = 6$, consider the subgroup $H = \{(1), (12)\}$ of order $|H| = 2$. The index of H in S_3 is:

$$[S_3 : H] = \frac{|S_3|}{|H|} = \frac{6}{2} = 3$$

This confirms our earlier calculation where we found exactly 3 distinct left cosets: $(1)H$, $(13)H$, and $(23)H$.

Consequences of Lagrange's Theorem

1. **Order of an Element Divides Group Order**:

For any element a in a finite group G, the order of a divides the order of G.

$$o(a) \text{ divides } |G|$$

Proof: Let $a \in G$, and let $H = \langle a \rangle$ be the cyclic subgroup generated by a. The order of this subgroup is equal to the order of the element:

$$|H| = o(a)$$

By Lagrange's theorem, $|H|$ divides $|G|$. Consequently, $o(a)$ divides $|G|$.

1. **Group Order Power Identity**:

If G is a finite group of order n, then for any element $a \in G$:

$$a^n = e$$

Proof: Let $m = o(a)$. By the previous consequence, m divides n, so there exists an integer k such that:

$$n = m \cdot k$$

Raising a to the power of n:

$$a^n = a^{m \cdot k} = (a^m)^k$$

Since $a^m = e$ by definition of element order:

$$a^n = e^k = e$$

1. **Groups of Prime Order are Cyclic**:

Every group of prime order is cyclic and has no non-trivial subgroups. *Proof*: Let G be a group of prime order $|G| = p$. Let $a \in G$ such that $a \neq e$. Consider the cyclic subgroup $H = \langle a \rangle$. By Lagrange's theorem, $|H|$ must divide $|G| = p$. Since p is prime, its only divisors are 1 and p. Because $a \neq e$, the subgroup H contains at least e and a, so $|H| > 1$. Therefore:

$$|H| = p$$

Since $H \subseteq G$ and $|H| = |G|$, we have $H = G$. Thus:

$$G = \langle a \rangle$$

Hence, G is cyclic, generated by any non-identity element $a \in G$.

4. Normal Subgroups and Factor Groups

Normal Subgroups

Let G be a group under a binary operation, and let H be a subgroup of G. For any element $g \in G$, the left coset of H determined by g is the set $gH = \{gh : h \in H\}$, and the right coset is $Hg = \{hg : h \in H\}$. In general, left and right cosets of a subgroup need not coincide. A subgroup for which left and right cosets are identical for every element of the group plays a central role in group structure.

A subgroup H of G is defined to be a normal subgroup of G, denoted by $H \triangleleft G$, if for every element $g \in G$, the left coset gH equals the right coset Hg.

$$gH = Hg \quad \forall g \in G$$

To test whether a subgroup H is normal in G, several equivalent criteria can be employed.

The primary test for normality is the element-wise conjugate test. A subgroup H is normal in G if and only if for every element $g \in G$ and every element $h \in H$, the conjugate ghg^{-1} belongs to H.

$$ghg^{-1} \in H \quad \forall g \in G,\ \forall h \in H$$

To derive this criterion from the definition, assume that $gH = Hg$ for all $g \in G$. For any element $h \in H$, the element gh belongs to the left coset gH. Since $gH = Hg$, the element gh must also belong to the right coset Hg. Consequently, there exists some element $h' \in H$ such that:

$$gh = h'g$$

Multiplying both sides of this equation on the right by the inverse element g^{-1} gives:

$$(gh)g^{-1} = (h'g)g^{-1}$$

Using the associativity of group multiplication, the right-hand side simplifies as:

$$ghg^{-1} = h'(gg^{-1})$$

Since $gg^{-1} = e$, where e is the identity element of G, we obtain:

$$ghg^{-1} = h'e$$

$$ghg^{-1} = h'$$

Because $h' \in H$, it follows directly that:

$$ghg^{-1} \in H$$

Conversely, if $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$, then multiplying on the right by g yields:

$$(ghg^{-1})g \in Hg$$

$$gh(g^{-1}g) \in Hg$$

$$gh \in Hg$$

This shows that $gH \subseteq Hg$. Replacing g with g^{-1} in the normality condition yields $g^{-1}h(g^{-1})^{-1} \in H$, which simplifies to $g^{-1}hg \in H$. Multiplying on the left by g yields:

$$g(g^{-1}hg) \in gH$$

$$(gg^{-1})hg \in gH$$

$$hg \in gH$$

This proves that $Hg \subseteq gH$. Combining both inclusions yields $gH = Hg$, confirming that the test is necessary and sufficient. An alternative equivalent form of the test expressed in terms of sets is:

$$gHg^{-1} = H \quad \forall g \in G$$

As a first fundamental example of normal subgroups, consider an abelian group G. In an abelian group, the group operation is commutative, meaning $gh = hg$ for all $g, h \in G$. Let H be any subgroup of an abelian group G. To check normality, consider the conjugate of an element $h \in H$ by any element $g \in G$:

$$ghg^{-1} = (gh)g^{-1}$$

Applying commutativity to the term gh yields:

$$ghg^{-1} = (hg)g^{-1}$$

By associativity, this becomes:

$$ghg^{-1} = h(gg^{-1})$$

$$ghg^{-1} = he$$

$$ghg^{-1} = h$$

Since $h \in H$, it follows that $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$. Therefore, every subgroup of an abelian group is a normal subgroup.

As a second standard example, consider the symmetric group S_n consisting of all permutations on n elements, and its subgroup A_n, the alternating group consisting of all even permutations. The order of S_n is $n!$, and the order of A_n is $\frac{n!}{2}$. The index of A_n in S_n, defined as the number of left cosets, is given by:

$$[S_n : A_n] = \frac{|S_n|}{|A_n|}$$

$$[S_n : A_n] = \frac{n!}{\frac{n!}{2}} = 2$$

Since there are only two left cosets of A_n in S_n, one coset is A_n itself (the set of even permutations), and the remaining left coset must consist of all odd permutations, represented as gA_n for any odd permutation $g \in S_n$. Similarly, the right cosets are A_n and $A_n g$ (the set of odd permutations). Thus, for any element $g \in S_n$: - If $g \in A_n$, then $gA_n = A_n = A_n g$. - If $g \notin A_n$, then $gA_n = S_n \setminus A_n = A_n g$.

In both cases, $gA_n = A_n g$, proving that the alternating group A_n is a normal subgroup of S_n.

Factor Groups

When H is a normal subgroup of G, the set of all cosets of H in G can itself be given the structure of a group. This group is called the factor group or quotient group of G by H, denoted by G/H.

The set underlying the factor group is defined as:

$$G/H = \{aH : a \in G\}$$

The operation on G/H is defined via coset multiplication:

$$(aH)(bH) = (ab)H \quad \forall a, b \in G$$

For this operation to be meaningful, it must be well-defined; that is, the product coset $(ab)H$ must not depend on the choice of representatives a and b selected from the cosets aH and bH.

To verify well-definedness, let $a', b' \in G$ be alternative representatives such that $a'H = aH$ and $b'H = bH$. This implies that there exist elements $h_1, h_2 \in H$ such that:

$$a' = ah_1$$

$$b' = bh_2$$

Now consider the product of the alternative representatives:

$$a'b' = (ah_1)(bh_2)$$

By associativity, this is written as:

$$a'b' = a(h_1 b)h_2$$

Because H is a normal subgroup, the left coset bH is equal to the right coset Hb. Since $h_1 \in H$, the element $h_1 b$ lies in $Hb = bH$. Therefore, there exists an element $h_3 \in H$ such that:

$$h_1 b = bh_3$$

Substituting this identity into the expression for $a'b'$ yields:

$$a'b' = a(bh_3)h_2$$

Using associativity, we regroup the terms as:

$$a'b' = (ab)(h_3h_2)$$

Since H is a subgroup closed under multiplication, the product h_3h_2 belongs to H. Let $h_4 = h_3h_2 \in H$, so that:

$$a'b' = (ab)h_4$$

Taking the coset generated by $a'b'$ gives:

$$(a'b')H = ((ab)h_4)H$$

Since $h_4 \in H$, the coset h_4H is simply H, which leads to:

$$(a'b')H = (ab)H$$

This completes the proof that coset multiplication is well-defined if and only if H is a normal subgroup of G.

The group axioms for G/H under this operation are satisfied as follows:

1. **Closure**: For any $aH, bH \in G/H$, the product $(aH)(bH) = (ab)H$ is a valid coset in G/H because $ab \in G$.
2. **Associativity**: For any $aH, bH, cH \in G/H$:

$$((aH)(bH))(cH) = ((ab)H)(cH) = ((ab)c)H$$

Using associativity in G, $(ab)c = a(bc)$, so:

$$((ab)c)H = (a(bc))H = (aH)((bc)H) = (aH)((bH)(cH))$$

1. **Identity Element**: The identity element of G/H is the coset $eH = H$, where e is the identity element of G. For any $aH \in G/H$:

$$(eH)(aH) = (ea)H = aH$$

$$(aH)(eH) = (ae)H = aH$$

1. **Inverse Element**: For any element $aH \in G/H$, the inverse element is $a^{-1}H$. This is verified by:

$$(aH)(a^{-1}H) = (aa^{-1})H = eH = H$$

$$(a^{-1}H)(aH) = (a^{-1}a)H = eH = H$$

As an example of a factor group, consider the additive group of integers $G = \mathbb{Z}$ and the subgroup $H = n\mathbb{Z}$ consisting of all multiples of a fixed integer n. Since $\mathbb{Z}$ is an abelian group, $n\mathbb{Z}$ is automatically normal in $\mathbb{Z}$. The factor group is:

$$\mathbb{Z}/n\mathbb{Z} = \{k + n\mathbb{Z} : k \in \{0, 1, 2, \ldots, n-1\}\}$$

The group operation in $\mathbb{Z}/n\mathbb{Z}$ is addition of cosets, defined as:

$$(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = (a + b) + n\mathbb{Z}$$

This factor group is isomorphic to the group of integers modulo n, denoted $\mathbb{Z}_n$.

5. Classes

Conjugacy Classes

Conjugacy defines an equivalence relation on the elements of a group G, allowing the group to be partitioned into disjoint subsets called conjugacy classes.

Two elements $a, b \in G$ are said to be conjugate to each other, written as $a \sim b$, if there exists an element $g \in G$ such that:

$$b = gag^{-1}$$

To confirm that conjugacy is an equivalence relation, we verify three properties:

1. **Reflexivity**: For any $a \in G$, choosing $g = e$ (the identity element) gives:

$$eae^{-1} = eae = a$$

Thus, $a \sim a$.

1. **Symmetry**: Suppose $a \sim b$, so that $b = gag^{-1}$ for some $g \in G$. Multiplying on the left by g^{-1} gives:

$$g^{-1}b = g^{-1}(gag^{-1})$$

$$g^{-1}b = (g^{-1}g)ag^{-1}$$

$$g^{-1}b = eag^{-1} = ag^{-1}$$

Multiplying on the right by g gives:

$$(g^{-1}b)g = (ag^{-1})g$$

$$g^{-1}bg = a(g^{-1}g)$$

$$g^{-1}bg = ae = a$$

Letting $k = g^{-1} \in G$, we write $k^{-1} = (g^{-1})^{-1} = g$, so:

$$a = kbk^{-1}$$

Thus, $b \sim a$.

1. **Transitivity**: Suppose $a \sim b$ and $b \sim c$. Then there exist elements $g, k \in G$ such that $b = gag^{-1}$ and $c = kbk^{-1}$. Substituting the expression for b into the equation for c:

$$c = k(gag^{-1})k^{-1}$$

Using associativity:

$$c = (kg)a(g^{-1}k^{-1})$$

Since $(kg)^{-1} = g^{-1}k^{-1}$, this becomes:

$$c = (kg)a(kg)^{-1}$$

Setting $x = kg \in G$, we have $c = xax^{-1}$, which proves $a \sim c$.

Since conjugacy is an equivalence relation, it partitions G into mutually disjoint equivalence classes. The conjugacy class of an element $a \in G$, denoted $C(a)$, is the set of all elements in G conjugate to a:

$$C(a) = \{gag^{-1} : g \in G\}$$

Because the conjugacy classes form a partition of G, the group G is the disjoint union of its distinct conjugacy classes:

$$G = \bigcup_i C_i$$

Taking the order (number of elements) of both sides gives the class equation idea for a finite group G:

$$|G| = \sum_i |C_i|$$

where the summation runs over one representative element from each distinct conjugacy class.

Centre of a Group

The centre of a group G, denoted by $Z(G)$, is defined as the set of all elements in G that commute with every element of G:

$$Z(G) = \{z \in G : zg = gz \quad \forall g \in G\}$$

The condition $zg = gz$ for all $g \in G$ can be rewritten by multiplying on the right by g^{-1}:

$$(zg)g^{-1} = (gz)g^{-1}$$

$$z(gg^{-1}) = gzg^{-1}$$

$$ze = gzg^{-1}$$

$$z = gzg^{-1} \quad \forall g \in G$$

This shows that an element z belongs to the centre $Z(G)$ if and only if its conjugate by any element $g \in G$ is z itself. Consequently, the conjugacy class of any central element $z \in Z(G)$ consists of only one element:

$$C(z) = \{z\}$$

Thus, $|C(z)| = 1$ if and only if $z \in Z(G)$.

We can split the class equation into two parts: the sum over elements belonging to the centre $Z(G)$ (each contributing a class of size 1), and the sum over representatives of non-central conjugacy classes (each having size strictly greater than 1).

Since there are $|Z(G)|$ elements in the centre, each producing a distinct conjugacy class of size 1, their contribution to the total order of the group is $|Z(G)| \times 1 = |Z(G)|$. The class equation can therefore be refined as:

$$|G| = |Z(G)| + \sum_i |C_i|$$

where the sum ranges over one representative for each distinct conjugacy class C_i with $|C_i| > 1$.

6. Direct Product of Groups

Let $(G, \cdot)$ and $(H, *)$ be two groups with identity elements e_G and e_H, respectively. The direct product of G and H, denoted by $G \times H$, is defined as the Cartesian product set of ordered pairs:

$$G \times H = \{(g, h) : g \in G, h \in H\}$$

The group operation on $G \times H$ is defined component-wise. For any two elements $(g_1, h_1), (g_2, h_2) \in G \times H$:

$$(g_1, h_1)(g_2, h_2) = (g_1 \cdot g_2, h_1 * h_2)$$

We verify that $G \times H$ satisfies all group axioms under component-wise multiplication:

1. **Closure**: For any $(g_1, h_1), (g_2, h_2) \in G \times H$, since $g_1 \cdot g_2 \in G$ and $h_1 * h_2 \in H$, the pair $(g_1 \cdot g_2, h_1 * h_2)$ belongs to $G \times H$.

1. **Associativity**: For any $(g_1, h_1), (g_2, h_2), (g_3, h_3) \in G \times H$:

$$((g_1, h_1)(g_2, h_2))(g_3, h_3) = (g_1 \cdot g_2, h_1 * h_2)(g_3, h_3)$$

$$(g_1 \cdot g_2, h_1 * h_2)(g_3, h_3) = ((g_1 \cdot g_2) \cdot g_3, (h_1 * h_2) * h_3)$$

Using associativity in G and H:

$$((g_1 \cdot g_2) \cdot g_3, (h_1 * h_2) * h_3) = (g_1 \cdot (g_2 \cdot g_3), h_1 * (h_2 * h_3))$$

$$(g_1 \cdot (g_2 \cdot g_3), h_1 * (h_2 * h_3)) = (g_1, h_1)(g_2 \cdot g_3, h_2 * h_3)$$

$$(g_1, h_1)(g_2 \cdot g_3, h_2 * h_3) = (g_1, h_1)((g_2, h_2)(g_3, h_3))$$

1. **Identity Element**: The identity element of $G \times H$ is $e_{G \times H} = (e_G, e_H)$. For any $(g, h) \in G \times H$:

$$(g, h)(e_G, e_H) = (g \cdot e_G, h * e_H) = (g, h)$$

$$(e_G, e_H)(g, h) = (e_G \cdot g, e_H * h) = (g, h)$$

1. **Inverse Element**: For any element $(g, h) \in G \times H$, its inverse is $(g^{-1}, h^{-1}) \in G \times H$, since:

$$(g, h)(g^{-1}, h^{-1}) = (g \cdot g^{-1}, h * h^{-1}) = (e_G, e_H)$$

$$(g^{-1}, h^{-1})(g, h) = (g^{-1} \cdot g, h^{-1} * h) = (e_G, e_H)$$

If G and H are finite groups, the order of the direct product group is the product of their individual orders:

$$|G \times H| = |G| \times |H|$$

As a concrete example, consider the direct product group $\mathbb{Z}_2 \times \mathbb{Z}_2$, where $\mathbb{Z}_2 = \{0, 1\}$ is the group of integers modulo 2 under addition. The elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$ are the four ordered pairs:

$$\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$$

The identity element is $(0, 0)$. Addition is defined component-wise modulo 2. For each non-identity element, we compute its sum with itself:

$$(1, 0) + (1, 0) = (1 + 1 \pmod 2, 0 + 0 \pmod 2) = (0, 0)$$

$$(0, 1) + (0, 1) = (0 + 0 \pmod 2, 1 + 1 \pmod 2) = (0, 0)$$

$$(1, 1) + (1, 1) = (1 + 1 \pmod 2, 1 + 1 \pmod 2) = (0, 0)$$

Each non-identity element has order 2, meaning $x + x = (0, 0)$ for all $x \in \mathbb{Z}_2 \times \mathbb{Z}_2$. This group structure is isomorphic to the Klein four-group, often denoted by V_4. The Klein four-group consists of four elements $V_4 = \{e, a, b, c\}$ with the identity e, where every element is its own inverse:

$$a^2 = b^2 = c^2 = e$$

and the product of any two distinct non-identity elements yields the third:

$$ab = ba = c$$

$$bc = cb = a$$

$$ca = ac = b$$

The explicit correspondence mapping $(0, 0) \mapsto e$, $(1, 0) \mapsto a$, $(0, 1) \mapsto b$, and $(1, 1) \mapsto c$ shows that $\mathbb{Z}_2 \times \mathbb{Z}_2$ is structurally identical to the Klein four-group V_4.

7. Homomorphism and Isomorphism

Definitions

Let $(G, \cdot)$ and $(H, *)$ be two groups. A group homomorphism is a mapping $\phi : G \to H$ that preserves the group operation. That is, for all elements $a, b \in G$:

$$\phi(a \cdot b) = \phi(a) * \phi(b)$$

From this definition, two fundamental properties follow directly:

1. **Preservation of Identity**: Let e_G be the identity of G and e_H be the identity of H. For any $a \in G$, we have $a \cdot e_G = a$. Applying ϕ:

$$\phi(a \cdot e_G) = \phi(a)$$

$$\phi(a) * \phi(e_G) = \phi(a)$$

Multiplying both sides on the left by $(\phi(a))^{-1}$ in H:

$$(\phi(a))^{-1} * (\phi(a) * \phi(e_G)) = (\phi(a))^{-1} * \phi(a)$$

$$((\phi(a))^{-1} * \phi(a)) * \phi(e_G) = e_H$$

$$e_H * \phi(e_G) = e_H$$

$$\phi(e_G) = e_H$$

1. **Preservation of Inverses**: For any $a \in G$, $a \cdot a^{-1} = e_G$. Applying ϕ:

$$\phi(a \cdot a^{-1}) = \phi(e_G)$$

$$\phi(a) * \phi(a^{-1}) = e_H$$

Multiplying on the left by $(\phi(a))^{-1}$ gives:

$$\phi(a^{-1}) = (\phi(a))^{-1}$$

A group isomorphism is a group homomorphism $\phi : G \to H$ that is bijective (both one-to-one and onto). If there exists a group isomorphism between G and H, the groups G and H are said to be isomorphic, written as $G \cong H$. Isomorphic groups share identical algebraic properties and differ only in the notation used for their elements.

Kernel and Image of a Homomorphism

For a homomorphism $\phi : G \to H$, two important sets are defined: the kernel and the image.

The kernel of ϕ, denoted $\ker(\phi)$, is the set of all elements in G that are mapped by ϕ to the identity element e_H of H:

$$\ker(\phi) = \{g \in G : \phi(g) = e_H\}$$

We now prove that $\ker(\phi)$ is always a normal subgroup of G.

First, to show that $\ker(\phi)$ is a subgroup of G: - **Non-emptiness**: Since $\phi(e_G) = e_H$, $e_G \in \ker(\phi)$. - **Closure under product and inverse**: Let $x, y \in \ker(\phi)$, so that $\phi(x) = e_H$ and $\phi(y) = e_H$. Consider $\phi(xy^{-1})$:

$$\phi(xy^{-1}) = \phi(x) * \phi(y^{-1})$$

$$\phi(xy^{-1}) = \phi(x) * (\phi(y))^{-1}$$

$$\phi(xy^{-1}) = e_H * (e_H)^{-1}$$

$$\phi(xy^{-1}) = e_H * e_H = e_H$$

Thus, $xy^{-1} \in \ker(\phi)$, confirming that $\ker(\phi)$ is a subgroup of G.

Second, to show that $\ker(\phi)$ is normal in G: Let $g \in G$ be any element and let $k \in \ker(\phi)$, so that $\phi(k) = e_H$. We compute $\phi(gkg^{-1})$:

$$\phi(gkg^{-1}) = \phi(g) * \phi(k) * \phi(g^{-1})$$

Substituting $\phi(k) = e_H$:

$$\phi(gkg^{-1}) = \phi(g) * e_H * \phi(g^{-1})$$

$$\phi(gkg^{-1}) = \phi(g) * \phi(g^{-1})$$

Using the homomorphic property:

$$\phi(gkg^{-1}) = \phi(gg^{-1})$$

$$\phi(gkg^{-1}) = \phi(e_G) = e_H$$

Therefore, $gkg^{-1} \in \ker(\phi)$ for all $g \in G$ and $k \in \ker(\phi)$, proving that $\ker(\phi) \triangleleft G$.

The image of ϕ, denoted $\mathrm{Im}(\phi)$ or $\phi(G)$, is the subset of H consisting of all values taken by ϕ:

$$\mathrm{Im}(\phi) = \{\phi(g) \in H : g \in G\}$$

The image $\mathrm{Im}(\phi)$ is a subgroup of H.

As an example of a homomorphism, consider $\phi : \mathbb{Z} \to \mathbb{Z}_n$ defined by $\phi(k) = k \pmod n$, where the operation on $\mathbb{Z}$ is standard addition and the operation on $\mathbb{Z}_n$ is addition modulo n. For any $a, b \in \mathbb{Z}$:

$$\phi(a+b) = (a+b) \pmod n = (a \pmod n + b \pmod n) \pmod n = \phi(a) + \phi(b)$$

The kernel of this homomorphism is the set of all integers mapping to $0 \pmod n$:

$$\ker(\phi) = \{k \in \mathbb{Z} : k \pmod n = 0\} = n\mathbb{Z}$$

The image of ϕ is the entire group $\mathbb{Z}_n$.

As a second example, consider the parity or sign homomorphism $\mathrm{sgn} : S_n \to (\{1, -1\}, \times)$ defined on the symmetric group S_n. For any permutation $\sigma \in S_n$, $\mathrm{sgn}(\sigma) = 1$ if σ is an even permutation and $\mathrm{sgn}(\sigma) = -1$ if σ is an odd permutation. For any permutations $\sigma, \tau \in S_n$:

$$\mathrm{sgn}(\sigma\tau) = \mathrm{sgn}(\sigma) \cdot \mathrm{sgn}(\tau)$$

The kernel of sgn is the set of permutations mapping to the identity element 1:

$$\ker(\mathrm{sgn}) = \{\sigma \in S_n : \mathrm{sgn}(\sigma) = 1\} = A_n$$

This gives an alternative proof that the alternating group A_n is a normal subgroup of S_n, since it is the kernel of a group homomorphism.

First Isomorphism Idea

The First Isomorphism Theorem provides a fundamental link between homomorphisms, factor groups, and images.

Let $\phi : G \to H$ be a group homomorphism with kernel $K = \ker(\phi)$ and image $\mathrm{Im}(\phi)$. The theorem states that the factor group G/K is isomorphic to the image $\mathrm{Im}(\phi)$:

$$G/\ker(\phi) \cong \mathrm{Im}(\phi)$$

To understand this result, define a mapping $\psi : G/K \to \mathrm{Im}(\phi)$ by:

$$\psi(gK) = \phi(g)$$

We check that ψ is well-defined. Suppose $g_1K = g_2K$. Then $g_2^{-1}g_1 \in K = \ker(\phi)$, which implies:

$$\phi(g_2^{-1}g_1) = e_H$$

$$\phi(g_2)^{-1} * \phi(g_1) = e_H$$

$$\phi(g_1) = \phi(g_2)$$

$$\psi(g_1K) = \psi(g_2K)$$

Thus ψ is well-defined. Furthermore, ψ is a homomorphism because:

$$\psi((g_1K)(g_2K)) = \psi((g_1g_2)K) = \phi(g_1g_2) = \phi(g_1) * \phi(g_2) = \psi(g_1K) * \psi(g_2K)$$

A key consequence of this structure is the trivial kernel criterion for injectivity: a homomorphism $\phi : G \to H$ is injective (one-to-one) if and only if its kernel is trivial, meaning $\ker(\phi) = \{e_G\}$.

To derive this criterion, assume first that ϕ is injective. If $k \in \ker(\phi)$, then $\phi(k) = e_H$. Since we also know $\phi(e_G) = e_H$, injectivity requires:

$$\phi(k) = \phi(e_G) \implies k = e_G$$

Thus $\ker(\phi) = \{e_G\}$.

Conversely, assume that $\ker(\phi) = \{e_G\}$. Suppose $\phi(a) = \phi(b)$ for some $a, b \in G$. Multiplying on the right by $(\phi(b))^{-1}$ gives:

$$\phi(a) * (\phi(b))^{-1} = \phi(b) * (\phi(b))^{-1}$$

$$\phi(a) * \phi(b^{-1}) = e_H$$

Applying the homomorphic property to the left-hand side:

$$\phi(ab^{-1}) = e_H$$

By definition of the kernel, this identity implies:

$$ab^{-1} \in \ker(\phi)$$

Since the kernel is trivial, $\ker(\phi) = \{e_G\}$, we must have:

$$ab^{-1} = e_G$$

Multiplying on the right by b yields:

$$(ab^{-1})b = e_G b$$

$$a(b^{-1}b) = b$$

$$ae_G = b$$

$$a = b$$

This proves that ϕ is one-to-one. Therefore, when the kernel of a homomorphism ϕ is trivial ($\ker(\phi) = \{e_G\}$), the mapping ϕ forms a bijection between G and its image $\mathrm{Im}(\phi)$, making ϕ an isomorphism of G onto $\mathrm{Im}(\phi)$.

8. Bisection Method

The Intermediate Value Idea

The bisection method is a simple numerical technique for finding a real root of a continuous algebraic or transcendental equation of the form $f(x) = 0$. The theoretical foundation of this method rests upon the Intermediate Value Theorem for continuous functions.

If a function $f(x)$ is continuous on a closed interval $[a, b]$, and the values of the function at the endpoints have opposite algebraic signs such that

$$f(a) \cdot f(b) < 0$$

then there exists at least one real value $\alpha \in (a, b)$ for which the function vanishes, satisfying

$$f(\alpha) = 0$$

This guarantees that the interval $[a, b]$ brackets at least one real root of the equation.

The Algorithm

The bisection algorithm systematically reduces the length of the interval containing the root by halving it at each iteration.

Given an initial interval $[a_0, b_0]$ where $f(a_0) \cdot f(b_0) < 0$, the first midpoint c_0 is computed by taking the arithmetic mean of the endpoints

$$c_0 = \frac{a_0 + b_0}{2}$$

The function value $f(c_0)$ at this midpoint is evaluated. If $f(c_0) = 0$, then c_0 is the exact root of the equation, and the iteration terminates immediately.

If $f(c_0) \neq 0$, the sign of $f(c_0)$ is compared with the signs of $f(a_0)$ and $f(b_0)$. The new sub-interval $[a_1, b_1]$ containing the root is selected such that the function values at its endpoints maintain opposite signs.

If the product of the function values at the left endpoint and the midpoint is negative, satisfying

$$f(a_0) \cdot f(c_0) < 0$$

the root lies within the left half of the interval. Consequently, the upper bound is updated while the lower bound remains unchanged, giving

$$a_1 = a_0$$

$$b_1 = c_0$$

If the product of the function values at the left endpoint and the midpoint is positive, satisfying

$$f(a_0) \cdot f(c_0) > 0$$

then by necessity the sign change occurs in the right half of the interval, satisfying $f(c_0) \cdot f(b_0) < 0$. Consequently, the lower bound is updated while the upper bound remains unchanged, giving

$$a_1 = c_0$$

$$b_1 = b_0$$

This procedure is repeated iteratively to generate a sequence of nested intervals $[a_n, b_n]$ such that each interval is half the length of the previous one, continuously trapping the real root within a shrinking domain.

Convergence

To analyze the convergence of the bisection method, consider the initial interval $[a_0, b_0]$ of length $L_0 = b_0 - a_0$. After one iteration, the length of the interval $[a_1, b_1]$ becomes

$$L_1 = \frac{b_0 - a_0}{2}$$

After two iterations, the length of the interval $[a_2, b_2]$ becomes

$$L_2 = \frac{b_0 - a_0}{2^2}$$

By induction, after n successive iterations, the length of the interval $[a_n, b_n]$ is given by

$$L_n = \frac{b_0 - a_0}{2^n}$$

Let α be the exact root situated inside the interval $[a_n, b_n]$, and let $c_n = \frac{a_n + b_n}{2}$ be the estimated root at the n-th step. The absolute error e_n at the n-th iteration is bounded by half the width of the interval $[a_n, b_n]$, giving

$$e_n = |c_n - \alpha| \leq \frac{b_n - a_n}{2}$$

Substituting the expression for the interval length after n iterations into the error bound yields

$$e_n \leq \frac{b_0 - a_0}{2^{n+1}}$$

Taking the limit of the error bound as the number of iterations n approaches infinity yields

$$\lim_{n \to \infty} e_n \leq \lim_{n \to \infty} \frac{b_0 - a_0}{2^{n+1}} = 0$$

This proves that the bisection method is globally convergent, meaning it is guaranteed to converge to a root provided the initial bracket $[a_0, b_0]$ satisfies $f(a_0) \cdot f(b_0) < 0$. Because the error bound is reduced by a factor of $\frac{1}{2}$ at each step, the method exhibits linear convergence with an asymptotic error constant of $\frac{1}{2}$.

Worked Example: Root of $x - \cos x = 0$

To find the positive real root of the transcendental equation $x - \cos x = 0$ using the bisection method, we define the continuous function

$$f(x) = x - \cos x$$

We first determine an initial interval $[a_0, b_0]$ where the function changes sign. Evaluating $f(x)$ at $x = 0$ gives

$$f(0) = 0 - \cos(0) = 0 - 1 = -1$$

Evaluating $f(x)$ at $x = 1$ gives

$$f(1) = 1 - \cos(1) \approx 1 - 0.540302 = 0.459698$$

Since $f(0) < 0$ and $f(1) > 0$, the product satisfies $f(0) \cdot f(1) < 0$, confirming that a positive root lies in the interval $[a_0, b_0] = [0, 1]$.

For the first iteration, the midpoint c_0 is calculated as

$$c_0 = \frac{0 + 1}{2} = 0.5$$

Evaluating the function at $c_0 = 0.5$ yields

$$f(0.5) = 0.5 - \cos(0.5) \approx 0.5 - 0.877583 = -0.377583$$

Since $f(0.5) < 0$ and $f(1) > 0$, the root lies in the sub-interval $[a_1, b_1] = [0.5, 1]$.

For the second iteration, the midpoint c_1 is calculated as

$$c_1 = \frac{0.5 + 1}{2} = 0.75$$

Evaluating the function at $c_1 = 0.75$ yields

$$f(0.75) = 0.75 - \cos(0.75) \approx 0.75 - 0.731689 = 0.018311$$

Since $f(0.5) < 0$ and $f(0.75) > 0$, the root lies in the sub-interval $[a_2, b_2] = [0.5, 0.75]$.

For the third iteration, the midpoint c_2 is calculated as

$$c_2 = \frac{0.5 + 0.75}{2} = 0.625$$

Evaluating the function at $c_2 = 0.625$ yields

$$f(0.625) = 0.625 - \cos(0.625) \approx 0.625 - 0.810965 = -0.185965$$

Since $f(0.625) < 0$ and $f(0.75) > 0$, the root lies in the sub-interval $[a_3, b_3] = [0.625, 0.75]$.

For the fourth iteration, the midpoint c_3 is calculated as

$$c_3 = \frac{0.625 + 0.75}{2} = 0.6875$$

Evaluating the function at $c_3 = 0.6875$ yields

$$f(0.6875) = 0.6875 - \cos(0.6875) \approx 0.6875 - 0.772882 = -0.085382$$

Since $f(0.6875) < 0$ and $f(0.75) > 0$, the root lies in the sub-interval $[a_4, b_4] = [0.6875, 0.75]$.

For the fifth iteration, the midpoint c_4 is calculated as

$$c_4 = \frac{0.6875 + 0.75}{2} = 0.71875$$

Evaluating the function at $c_4 = 0.71875$ yields

$$f(0.71875) = 0.71875 - \cos(0.71875) \approx 0.71875 - 0.752406 = -0.033656$$

Since $f(0.71875) < 0$ and $f(0.75) > 0$, the root lies in the sub-interval $[a_5, b_5] = [0.71875, 0.75]$.

For the sixth iteration, the midpoint c_5 is calculated as

$$c_5 = \frac{0.71875 + 0.75}{2} = 0.734375$$

Evaluating the function at $c_5 = 0.734375$ yields

$$f(0.734375) = 0.734375 - \cos(0.734375) \approx 0.734375 - 0.742171 = -0.007796$$

Since $f(0.734375) < 0$ and $f(0.75) > 0$, the root lies in the sub-interval $[a_6, b_6] = [0.734375, 0.75]$.

For the seventh iteration, the midpoint c_6 is calculated as

$$c_6 = \frac{0.734375 + 0.75}{2} = 0.7421875$$

Evaluating the function at $c_6 = 0.7421875$ yields

$$f(0.7421875) = 0.7421875 - \cos(0.7421875) \approx 0.7421875 - 0.736959 = 0.005228$$

Since $f(0.734375) < 0$ and $f(0.7421875) > 0$, the root is bounded within $[0.734375, 0.7421875]$. Taking the average of these bounds gives the positive real root approximation as

$$x \approx 0.739$$

9. Newton-Raphson Method

Derivation from the Tangent Line

The Newton-Raphson method is an open iterative numerical technique used to find the real roots of a continuously differentiable function $f(x) = 0$. The geometric foundation of the method relies on approximating the curve $y = f(x)$ by its tangent line at a point near the root.

Let x_n be an initial or current approximation to the true root α of the equation $f(x) = 0$. The point on the curve corresponding to this approximation is $(x_n, f(x_n))$, and the slope of the curve at this point is given by the first derivative $f'(x_n)$.

The point-slope equation of the tangent line to the curve $y = f(x)$ at the point $(x_n, f(x_n))$ is written as

$$y - f(x_n) = f'(x_n)(x - x_n)$$

The next approximation x_{n+1} to the root is defined as the x-intercept of this tangent line, where $y = 0$. Setting $y = 0$ and $x = x_{n+1}$ in the tangent line equation gives

$$0 - f(x_n) = f'(x_n)(x_{n+1} - x_n)$$

Simplifying the left-hand side of the equation yields

$$-f(x_n) = f'(x_n)(x_{n+1} - x_n)$$

Assuming that the first derivative is non-zero, $f'(x_n) \neq 0$, dividing both sides of the equation by $f'(x_n)$ yields

$$\frac{-f(x_n)}{f'(x_n)} = x_{n+1} - x_n$$

Adding x_n to both sides isolates the updated approximation x_{n+1}, establishing the standard Newton-Raphson iterative formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Convergence Analysis

To establish the order of convergence of the Newton-Raphson method, we analyze how the error evolves between successive iterations. Let α be the exact root satisfying $f(\alpha) = 0$. The error e_n at the n-th iteration is defined as

$$e_n = x_n - \alpha$$

Rearranging this expression expresses x_n in terms of the true root and the error

$$x_n = \alpha + e_n$$

Similarly, the error at the $(n+1)$-th iteration is defined as

$$e_{n+1} = x_{n+1} - \alpha$$

Substituting the Newton-Raphson formula for x_{n+1} into the error expression yields

$$e_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} - \alpha$$

Replacing $x_n - \alpha$ with e_n simplifies the expression to

$$e_{n+1} = e_n - \frac{f(x_n)}{f'(x_n)}$$

Putting this equation over a common denominator gives

$$e_{n+1} = \frac{e_n f'(x_n) - f(x_n)}{f'(x_n)}$$

Assuming $f(x)$ is twice continuously differentiable, we expand $f(\alpha) = f(x_n - e_n)$ using a Taylor series expansion about x_n

$$f(\alpha) = f(x_n) - e_n f'(x_n) + \frac{e_n^2}{2} f''(x_n) - \mathcal{O}(e_n^3)$$

Since α is the exact root, $f(\alpha) = 0$, giving

$$0 = f(x_n) - e_n f'(x_n) + \frac{e_n^2}{2} f''(x_n) - \mathcal{O}(e_n^3)$$

Rearranging this Taylor expansion to isolate $e_n f'(x_n) - f(x_n)$ gives

$$e_n f'(x_n) - f(x_n) = \frac{e_n^2}{2} f''(x_n) - \mathcal{O}(e_n^3)$$

Substituting this expression into the numerator of the error equation yields

$$e_{n+1} = \frac{\frac{e_n^2}{2} f''(x_n) - \mathcal{O}(e_n^3)}{f'(x_n)}$$

Neglecting higher-order terms of e_n and evaluating the derivatives at the true root α as $x_n \to \alpha$ yields the asymptotic relation

$$e_{n+1} \approx \left(\frac{f''(\alpha)}{2f'(\alpha)} \right) e_n^2$$

This relation demonstrates that the error at step $n+1$ is proportional to the square of the error at step n. Thus, when the Newton-Raphson method converges to a simple root where $f'(\alpha) \neq 0$, it exhibits quadratic convergence with an order of convergence $p = 2$.

Worked Example 1: Real Root of $x^3 - 9x + 1 = 0$

To find a real root of the algebraic equation $x^3 - 9x + 1 = 0$, we define the function

$$f(x) = x^3 - 9x + 1$$

Differentiating $f(x)$ with respect to x yields the derivative

$$f'(x) = 3x^2 - 9$$

Substituting $f(x)$ and $f'(x)$ into the Newton-Raphson iterative formula gives

$$x_{n+1} = x_n - \frac{x_n^3 - 9x_n + 1}{3x_n^2 - 9}$$

We test initial values to locate a root. Evaluating $f(0)$ gives

$$f(0) = 0^3 - 9(0) + 1 = 1 > 0$$

Evaluating $f(1)$ gives

$$f(1) = 1^3 - 9(1) + 1 = -7 < 0$$

Since $f(0) > 0$ and $f(1) < 0$, a real root lies between 0 and 1. We choose the initial approximation

$$x_0 = 0$$

For the first iteration, we calculate $f(x_0)$ and $f'(x_0)$ at $x_0 = 0$:

$$f(0) = 0^3 - 9(0) + 1 = 1$$

$$f'(0) = 3(0)^2 - 9 = -9$$

Substituting these values into the recurrence relation yields

$$x_1 = 0 - \frac{1}{-9} = \frac{1}{9} \approx 0.111111$$

For the second iteration, we calculate $f(x_1)$ and $f'(x_1)$ at $x_1 = 0.111111$:

$$f(0.111111) = (0.111111)^3 - 9(0.111111) + 1 \approx 0.001372 - 1.000000 + 1 = 0.001372$$

$$f'(0.111111) = 3(0.111111)^2 - 9 \approx 0.037037 - 9 = -8.962963$$

Substituting these values into the recurrence relation yields

$$x_2 = 0.111111 - \frac{0.001372}{-8.962963} \approx 0.111111 + 0.000153 = 0.111264$$

For the third iteration, we calculate $f(x_2)$ and $f'(x_2)$ at $x_2 = 0.111264$:

$$f(0.111264) = (0.111264)^3 - 9(0.111264) + 1 \approx 0.001378 - 1.001376 + 1 = 0.000002$$

$$f'(0.111264) = 3(0.111264)^2 - 9 \approx 0.037139 - 9 = -8.962861$$

Substituting these values into the recurrence relation yields

$$x_3 = 0.111264 - \frac{0.000002}{-8.962861} \approx 0.111264$$

Since x_2 and x_3 match to six decimal places, the real root of $x^3 - 9x + 1 = 0$ near origin is

$$x \approx 0.111264$$

Worked Example 2: Real Root of $x^3 - 3x + 1 = 0$ between 1 and 2

To find the real root of the algebraic equation $x^3 - 3x + 1 = 0$ situated in the interval $(1, 2)$ correct up to three decimal places, we define the continuous function

$$f(x) = x^3 - 3x + 1$$

Differentiating $f(x)$ with respect to x yields

$$f'(x) = 3x^2 - 3$$

Substituting $f(x)$ and $f'(x)$ into the Newton-Raphson formula gives

$$x_{n+1} = x_n - \frac{x_n^3 - 3x_n + 1}{3x_n^2 - 3}$$

We check the function values at the endpoints of the interval $[1, 2]$. Evaluating at $x = 1$ gives

$$f(1) = 1^3 - 3(1) + 1 = -1$$

Evaluating at $x = 2$ gives

$$f(2) = 2^3 - 3(2) + 1 = 3$$

Since $f(1) < 0$ and $f(2) > 0$, a real root is confirmed to lie between 1 and 2. We select an initial guess at the midpoint of the interval

$$x_0 = 1.5$$

For the first iteration, we calculate $f(x_0)$ and $f'(x_0)$ at $x_0 = 1.5$:

$$f(1.5) = (1.5)^3 - 3(1.5) + 1 = 3.375 - 4.5 + 1 = -0.125$$

$$f'(1.5) = 3(1.5)^2 - 3 = 3(2.25) - 3 = 6.75 - 3 = 3.75$$

Substituting these values into the Newton-Raphson iterative formula yields

$$x_1 = 1.5 - \frac{-0.125}{3.75} = 1.5 + 0.033333 = 1.533333$$

For the second iteration, we calculate $f(x_1)$ and $f'(x_1)$ at $x_1 = 1.533333$:

$$f(1.533333) = (1.533333)^3 - 3(1.533333) + 1 \approx 3.604741 - 4.600000 + 1 = 0.004741$$

$$f'(1.533333) = 3(1.533333)^2 - 3 \approx 3(2.351110) - 3 = 7.053330 - 3 = 4.053330$$

Substituting these values into the Newton-Raphson iterative formula yields

$$x_2 = 1.533333 - \frac{0.004741}{4.053330} \approx 1.533333 - 0.001170 = 1.532163$$

For the third iteration, we calculate $f(x_2)$ and $f'(x_2)$ at $x_2 = 1.532163$:

$$f(1.532163) = (1.532163)^3 - 3(1.532163) + 1 \approx 3.596468 - 4.596489 + 1 = -0.000021$$

$$f'(1.532163) = 3(1.532163)^2 - 3 \approx 3(2.347523) - 3 = 7.042569 - 3 = 4.042569$$

Substituting these values into the Newton-Raphson iterative formula yields

$$x_3 = 1.532163 - \frac{-0.000021}{4.042569} \approx 1.532163 + 0.000005 = 1.532168$$

Rounding the value of x_3 to three decimal places gives 1.532. Thus, the real root of $x^3 - 3x + 1 = 0$ lying between 1 and 2, correct up to three decimal places, is

$$x \approx 1.532$$

10. Euler Method

The numerical solution of a first-order initial value problem involves finding approximate values of an unknown function $y(x)$ satisfying the differential equation

$$\frac{dy}{dx} = f(x, y)$$

subject to the initial condition $y(x_0) = y_0$. We seek the solution at discrete grid points $x_n = x_0 + nh$, where h is a fixed positive step size and $n = 0, 1, 2, \ldots$.

To derive Euler's method, we consider the Taylor series expansion of the exact function $y(x)$ evaluated at $x_{n+1} = x_n + h$ around the point x_n:

$$y(x_{n+1}) = y(x_n) + hy'(x_n) + \frac{h^2}{2!}y''(x_n) + \frac{h^3}{3!}y'''(x_n) + \ldots$$

Substituting the governing differential equation $y'(x_n) = f(x_n, y(x_n))$ into the Taylor series expansion yields

$$y(x_{n+1}) = y(x_n) + hf(x_n, y(x_n)) + \frac{h^2}{2}y''(\xi_n)$$

where $\xi_n \in (x_n, x_{n+1})$ represents the remainder term evaluated at an intermediate point. Truncating the series after the first derivative term gives the geometric interpretation of moving along the tangent line evaluated at (x_n, y_n). This defines the Euler iteration formula:

$$y_{n+1} = y_n + hf(x_n, y_n)$$

The accuracy of a numerical scheme is characterized by its truncation error. The local truncation error (LTE) is defined as the error introduced in a single step assuming the previous value $y_n = y(x_n)$ is exact. Subtracting the Euler formula from the exact Taylor series expansion gives

$$\text{LTE} = y(x_{n+1}) - y_{n+1} = \frac{h^2}{2}y''(\xi_n)$$

Because the leading omitted term is proportional to h^2, the local truncation error of Euler's method is of order h^2, written as

$$\text{LTE} = \mathcal{O}(h^2)$$

The global truncation error (GTE) is the cumulative error accumulated over an integration interval $[x_0, X]$. The number of steps N required to reach X from x_0 with step size h is given by

$$N = \frac{X - x_0}{h}$$

Since the total accumulated error after N steps is bounded by N times the maximum local truncation error, we evaluate the order of the global error as

$$\text{GTE} \sim N \times \text{LTE} = \left(\frac{X - x_0}{h} \right) \mathcal{O}(h^2) = \mathcal{O}(h)$$

Thus, Euler's method is a first-order accurate method, meaning that halving the step size h reduces the global error by approximately a factor of two.

11. Second-Order Runge-Kutta Method

While Euler's method requires only one function evaluation per step, its first-order accuracy demands a very small step size h for high precision. The second-order Runge-Kutta method improves accuracy by matching terms up to second order in the Taylor series without requiring the evaluation of higher-order analytical derivatives of $f(x, y)$.

To derive the midpoint form of the second-order Runge-Kutta method, we first express the exact Taylor series expansion of $y(x_n + h)$ up to second order:

$$y(x_n + h) = y(x_n) + hy'(x_n) + \frac{h^2}{2} y''(x_n) + \mathcal{O}(h^3)$$

Using the total derivative of $f(x, y(x))$ with respect to x via the chain rule, the second derivative $y''(x)$ is written as

$$y''(x) = \frac{d}{dx} f(x, y) = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} = f_x + f_y f$$

Substituting $y' = f$ and $y'' = f_x + f_y f$ into the Taylor expansion yields

$$y(x_n + h) = y(x_n) + hf(x_n, y_n) + \frac{h^2}{2} \left[f_x(x_n, y_n) + f_y(x_n, y_n) f(x_n, y_n) \right] + \mathcal{O}(h^3)$$

We now formulate an approximation for y_{n+1} using a slope evaluation at the midpoint of the step:

$$y_{n+1} = y_n + hk_2$$

where k_1 is the initial slope evaluated at (x_n, y_n):

$$k_1 = f(x_n, y_n)$$

and k_2 is the slope evaluated at the estimated midpoint $(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_1)$:

$$k_2 = f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_1 \right)$$

Expanding k_2 in a bivariate Taylor series about the point (x_n, y_n) gives

$$k_2 = f(x_n, y_n) + \frac{h}{2} f_x(x_n, y_n) + \frac{h}{2} k_1 f_y(x_n, y_n) + \mathcal{O}(h^2)$$

Substituting $k_1 = f(x_n, y_n)$ into this expansion yields

$$k_2 = f(x_n, y_n) + \frac{h}{2} f_x(x_n, y_n) + \frac{h}{2} f(x_n, y_n) f_y(x_n, y_n) + \mathcal{O}(h^2)$$

Multiplying k_2 by h produces

$$hk_2 = hf(x_n, y_n) + \frac{h^2}{2} \left[f_x(x_n, y_n) + f_y(x_n, y_n) f(x_n, y_n) \right] + \mathcal{O}(h^3)$$

Comparing this directly with the exact Taylor expansion shows that $y_n + hk_2$ matches the exact solution up to terms of order h^2. Consequently, the midpoint second-order Runge-Kutta updating formula is defined as

$$y_{n+1} = y_n + hf\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} f(x_n, y_n) \right)$$

The local truncation error of the second-order Runge-Kutta method is determined by the first omitted term in the Taylor series expansion:

$$\text{LTE} = y(x_{n+1}) - y_{n+1} = \mathcal{O}(h^3)$$

Accumulating this error over $N = (X - x_0)/h$ steps gives the global truncation error:

$$\text{GTE} \sim N \times \text{LTE} = \left(\frac{X - x_0}{h} \right) \mathcal{O}(h^3) = \mathcal{O}(h^2)$$

The second-order Runge-Kutta method is significantly more accurate than Euler's method because Euler's method evaluates the slope strictly at the initial point of each sub-interval, completely ignoring the curvature of the solution curve across $[x_n, x_{n+1}]$. By evaluating the slope at the midpoint of the interval, the Runge-Kutta method cancels out the leading $\mathcal{O}(h^2)$ error term, reducing the global error from $\mathcal{O}(h)$ to $\mathcal{O}(h^2)$.

12. Worked Example: Euler and Second-Order Runge-Kutta

Consider the initial value problem

$$\frac{dy}{dx} = x + y, \quad y(0) = 1$$

We wish to compute approximate values of $y(x)$ at $x = 0.1$ and $x = 0.2$ using a step size of $h = 0.1$ using both Euler's method and the midpoint second-order Runge-Kutta method, and compare them with the exact solution

$$y(x) = 2e^x - x - 1$$

Evaluating the exact analytical solution at the target points gives:

$$y(0.1) = 2e^{0.1} - 0.1 - 1 = 2(1.105171) - 1.1 = 1.110342$$

$$y(0.2) = 2e^{0.2} - 0.2 - 1 = 2(1.221403) - 1.2 = 1.242806$$

Calculations using the Euler Method

For the first step from $x_0 = 0$ to $x_1 = 0.1$ with initial value $y_0 = 1$: The derivative function at (x_0, y_0) is

$$f(x_0, y_0) = 0 + 1 = 1.000000$$

Applying the Euler update formula gives

$$y_1 = y_0 + hf(x_0, y_0) = 1 + 0.1(1.000000) = 1.100000$$

For the second step from $x_1 = 0.1$ to $x_2 = 0.2$ with $y_1 = 1.100000$: The derivative function at (x_1, y_1) is

$$f(x_1, y_1) = 0.1 + 1.100000 = 1.200000$$

Applying the Euler update formula gives

$$y_2 = y_1 + hf(x_1, y_1) = 1.100000 + 0.1(1.200000) = 1.220000$$

Calculations using the Second-Order Runge-Kutta Method

For the first step from $x_0 = 0$ to $x_1 = 0.1$ with $y_0 = 1$: The initial slope k_1 is

$$k_1 = f(x_0, y_0) = 0 + 1 = 1.000000$$

The coordinates of the midpoint are calculated as

$$x_0 + \frac{h}{2} = 0 + 0.05 = 0.05$$

$$y_0 + \frac{h}{2}k_1 = 1 + 0.05(1.000000) = 1.050000$$

The slope at the midpoint k_2 is

$$k_2 = f(0.05, 1.050000) = 0.05 + 1.050000 = 1.100000$$

Applying the second-order Runge-Kutta update formula gives

$$y_1 = y_0 + hk_2 = 1 + 0.1(1.100000) = 1.110000$$

For the second step from $x_1 = 0.1$ to $x_2 = 0.2$ with $y_1 = 1.110000$: The initial slope k_1 is

$$k_1 = f(x_1, y_1) = 0.1 + 1.110000 = 1.210000$$

The coordinates of the midpoint are calculated as

$$x_1 + \frac{h}{2} = 0.1 + 0.05 = 0.15$$

$$y_1 + \frac{h}{2}k_1 = 1.110000 + 0.05(1.210000) = 1.170500$$

The slope at the midpoint k_2 is

$$k_2 = f(0.15, 1.170500) = 0.15 + 1.170500 = 1.320500$$

Applying the second-order Runge-Kutta update formula gives

$$y_2 = y_1 + hk_2 = 1.110000 + 0.1(1.320500) = 1.242050$$

Comparison of Results

A comparison of the numerical approximations against the exact analytical solution shows the absolute errors for both methods:

At $x = 0.1$: - Exact Solution: $y(0.1) = 1.110342$ - Euler Method: $y_1 = 1.100000$, Absolute Error = $|1.110342 - 1.100000| = 0.010342$ - Second-Order Runge-Kutta: $y_1 = 1.110000$, Absolute Error = $|1.110342 - 1.110000| = 0.000342$

At $x = 0.2$: - Exact Solution: $y(0.2) = 1.242806$ - Euler Method: $y_2 = 1.220000$, Absolute Error = $|1.242806 - 1.220000| = 0.022806$ - Second-Order Runge-Kutta: $y_2 = 1.242050$, Absolute Error = $|1.242806 - 1.242050| = 0.000756$

The second-order Runge-Kutta method yields results that are substantially closer to the exact solution than Euler's method, reducing the absolute error at $x = 0.2$ by a factor of over 30. This demonstrates the clear practical advantage of using a second-order scheme over a first-order scheme.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **There are 3 Assam University questions for this unit in the paper bank**, from PHYDSC-101T 1st Sem 2023, marked [AU] with their real source. The long-form questions are from other sources, marked as such. Note: there are no AU questions on group theory, and no AU questions on Euler or Runge-Kutta methods.

Short-form questions

S1. [AU] Explain bisection method of solving differential equation. — PHYDSC-101T, 1st Sem, 2023, FYUG (Sec B, 4 marks). Note: the paper says "differential equation" but the bisection method solves algebraic and transcendental equations, not differential equations.

Answer in brief: Section 8 — find an interval [a, b] with f(a) and f(b) of opposite signs, halve it repeatedly, keeping the half where the sign changes; the root is bracketed and the interval shrinks by half each step.

S2. [AU] Find the real root of the equation x^3 - 9x + 1 = 0. — PHYDSC-101T, 1st Sem, 2023, FYUG (Sec B, 4 marks) (Newton-Raphson; follows the bisection question).

Answer in brief: Section 9 — use $x_{n+1} = x_n - f(x_n)/f'(x_n)$ with $f(x) = x^3 - 9x + 1$; the worked example in the notes finds a root near $x = 0.111$.

S3. [AU] Write down Simpson's rule for integration of a function. — PHYDSC-101T, 1st Sem, 2023, FYUG (Sec A). Note: Simpson's rule is not in this unit's syllabus and is not covered in these notes.

Answer in brief: Not covered in these notes — Simpson's rule is a numerical integration formula and is outside the Unit 5 syllabus (which covers bisection, Newton-Raphson, Euler and Runge-Kutta methods only).

S4. [standard] Define a group and give two examples.

Answer in brief: Section 1 — a set with an operation that is closed, associative, has an identity element, and every element has an inverse; e.g. integers under addition, non-zero reals under multiplication.

S5. [standard] Define subgroup, coset and normal subgroup.

Answer in brief: Sections 2–4 — a subgroup is a subset that is itself a group; a (left) coset is $gH = \{gh : h \in H\}$; a subgroup is normal if $gH = Hg$ for every g.

S6. [standard] State Lagrange's theorem.

Answer in brief: Section 3 — the order of a subgroup divides the order of the (finite) group; equivalently $|G| = |H| \cdot [G : H]$.

S7. [standard] Define homomorphism and isomorphism.

Answer in brief: Section 7 — a homomorphism preserves the operation, $\phi(ab) = \phi(a)\phi(b)$; an isomorphism is a bijective homomorphism.

S8. [standard] State the Newton-Raphson iteration formula.

Answer in brief: Section 9 — $x_{n+1} = x_n - f(x_n)/f'(x_n)$.

S9. [standard] State the Euler method formula for dy/dx = f(x, y).

Answer in brief: Section 10 — $y_{n+1} = y_n + h\, f(x_n, y_n)$ with step h.

S10. [standard] State the second-order Runge-Kutta (midpoint) formula.

Answer in brief: Section 11 — $y_{n+1} = y_n + h\, f(x_n + h/2,\ y_n + (h/2) f(x_n, y_n))$.

Long-form questions

L1. [non-AU] Find the real root of x^3-3x+1=0 lying between 1 and 2 up to three decimal places by Newton Raphson method. — "Numerical analysis and statistics" Indian question bank, Section B.

What to write: Section 9 in full — iterate $x_{n+1} = x_n - (x_n^3 - 3x_n + 1)/(3x_n^2 - 3)$ from a starting value in [1, 2] until the value is stable to three decimal places (the root is near 1.532).

L2. [non-AU] Find the positive root of x - cos x = 0 by bisection method. — "Numerical analysis and statistics" Indian question bank, Section C.

What to write: Section 8 in full — bracket the root (f(0) = -1, f(1) > 0), halve the interval step by step, keeping the half where the sign changes, until the interval is small enough (the root is near 0.739).

L3. [standard] Define subgroup, coset and normal subgroup; state and prove Lagrange's theorem, show when a subgroup is normal, and give an example of a factor group and of a homomorphism and an isomorphism.

What to write: Sections 2–7 in full — definitions with examples; Lagrange's theorem from the coset partition; normality test $gHg^{-1} = H$; a factor group such as $\mathbb{Z}/n\mathbb{Z}$; a homomorphism such as the determinant map, and an isomorphism such as $\mathbb{R}$ under addition to positive reals under multiplication via the exponential.

L4. [standard] Solve dy/dx = x + y, y(0) = 1, for x = 0.1 and x = 0.2 by Euler method and by 2nd order Runge-Kutta method, and compare the two results.

What to write: Sections 10–12 in full — Euler gives y(0.1) = 1.1, y(0.2) = 1.22; second-order Runge-Kutta gives y(0.1) = 1.11, y(0.2) = 1.2421; the exact values are y(0.1) = 1.1103, y(0.2) = 1.2428, so Runge-Kutta is far closer.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.